

CODA → Master Planner: Handoff Report (Army / Talent Domain)

Agent: CODA (Cursor agent; Army officer talent / OER / Cox-competing-risks research)

Domain root: `./talent/` (canonical code: `talent/talent_pipeline/`; symlink at repo root: `talent_pipeline/` → same tree)

Report date: 2026-06-08 (refresh for Master Planner feed-in)

Prepared for: Charles Levine → future **Master Planner** agent (Cursor)

Charles's explicit instruction (2026-06-08): This report feeds the Master Planner. **Do not** begin drafting the cross-project master plan until Charles **assigns** that agent and asks for planning work. CODA/SCOUT/PEER/VECTOR should only **report status** until then.

0. What this document is (and is not)

What it is

- A **ground-truth status snapshot** of the Army/talent lane as CODA understands it after the **April 2026** thread (AWS sync, TB-stratified CR add-on, pool-size concerns, documentation).
- A **research narrative** (§2) for a Master Planner with **zero prior context**: ranks, promotions, pool quality, competing risks, Cox models, and the **inverted-U** finding — plus how Army links to SCOUT and PEER.
- A **file map** so the Master Planner can navigate without re-deriving context from chat logs.
- A **list of open questions** where Charles's intent is not fully locked — CODA prefers explicit answers over assumptions.
- Pointers to **SCOUT** (basketball), **PEER** (tenure), and **VECTOR** (manuscript) without claiming authority over those domains.

What it is not

- **Not** the master plan itself.
 - **Not** a commit to merge Army code paths with sports/tenure implementations (those are parallel settings with shared *ideas*, different estimands and data).
 - **Not** a guarantee that every local file has been pushed to Army AWS live — Charles transcribes/uploads manually; see §4.
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1. Thesis frame (why CODA exists in this repo)

Charles's dissertation uses **three empirical settings** under one mechanism story:

Setting	Agent (Charles’s naming)	Repo home	Outcome / mobility	Pool-quality construct
1 — Army	CODA	talent/	Promotion to Major vs attrition (competing risks)	Senior-rater / OER pool minus mean (LOO-style), binned for CIF panels
2 — Basketball	SCOUT	sports/	NBA draft (ever-draft v0, etc.)	LOO teammate pool quality (poolq_loo , generative extensions)
3 — Academia	PEER	tenure/	Tenure (Asst → Assoc)	LOO department prestige / pub intensity bins

CODA’s job: Make the Army pipeline **correct, documented, advisor-ready**, and **honest about estimands** (what a CIF bar *means* vs what a Cox HR *means*). The inverted-U / “middle pool wins, elite pool dips” narrative is the **qualitative target** linking all three settings; Army is the **most mature** quantitative stack (520 Cox + Cell 11 plotting).

VECTOR’s job (for Master Planner awareness): Turn cross-setting evidence into manuscript prose (1- Various_PDE_and_Chat_stuff/5-Manuscript/ — e.g. Vector_to_Scout_Tier1_Modeling_Direction.md , PEER_Status_Update_for_VECTOR_2026-06-03.md , Dakota committee brief). CODA supplies Army methods/results bullets; does not own VECTOR’s outline.

2. Army research program — zero-background primer

This section is for a Master Planner who has never seen the Army data or Charles’s dissertation thread. Status, files, and open tasks are in §3 onward.

2.1 Research question (plain English)

Charles studies whether **the quality of your peer rating pool** — the average performance of other officers evaluated by the same senior rater around the same time — predicts **upward mobility** (promotion) in a hierarchical performance system. The surprise is not a simple “better pool → always better outcomes” story. The empirical signature is an **inverted U**: promotion probability **rises** as pool quality increases through the middle of the distribution, then **falls** at the very highest pool-quality tiers (“elite pool” dip). That pattern is the **qualitative target** linking Army (Setting 1), college basketball draft (Setting 2, SCOUT), and academic tenure (Setting 3, PEER).

Mechanism hypotheses (still being articulated — see Publication_Plan.md): scarce promotion slots, congestion among high performers, signal dilution when everyone in the pool is strong, or comparative standing effects. Army is the **most mature** quantitative implementation; basketball has **replicated** the shape; tenure is **preliminary**.

2.2 Setting, ranks, and outcomes

Element	Army implementation
Population	U.S. Army officers, typically Combat Support / Combat Service Support cohorts in year groups such as 2002–2012 (run-specific filters in <code>pipeline_config*.py</code>)
Rank anchor	Observation begins after promotion to Captain (CPT) ; time-to-event is measured from that career anchor
Primary mobility outcome	Promotion to Major (MAJ)
Competing outcome	Attrition (separation before MAJ promotion — officers who attrite are no longer at risk for promotion)
Performance signal	Officer Evaluation Reports (OERs); top block (TB) share — how often an officer receives the highest rating category
Pool construct	Senior-rater pool : officers sharing the same senior rater (and snapshot timing) form a reference peer set. Pool quality $\approx$ mean TB of peers in that pool, computed leave-one-out (officer’s own rating excluded from the pool mean Charles calls “pool minus mean”)
Data	Proprietary Army personnel snapshots + OER records; cannot be shared publicly . Pipeline: 502 (snapshots) $\rightarrow$ 512 (OER assignment) $\rightarrow$ 520 (pool metrics, survival, plots, Cox)

Why ranks matter: The dissertation is about **advancement under constrained distinction** — who gets promoted when slots are scarce and everyone is rated relative to peers. CPT→MAJ is the canonical Army mobility run; attrition is treated as a **competing event**, not ignored censoring.

2.3 Two analysis layers (do not conflate them)

Charles uses **two complementary tools** on the same underlying panel. Master Planner and VECTOR must keep estimands distinct in manuscript language.

Layer A — Cell 11: Competing-risks descriptive plots (signature figures)

- **What:** Nonparametric **cumulative incidence function (CIF)** curves and **CIF bar panels** (e.g. **Q1–Q8** ordered bins on pool quality).
- **Y-axis (promotion bars):** **Final cumulative incidence** of promotion to MAJ — the share of officers in each pool-quality bin who have been promoted by end of follow-up, treating attrition as a competing risk in the display.
- **Signature finding:** Bars **rise** through mid bins, then **drop** in top bins $\rightarrow$ **inverted U** / elite-tier dip. Holds with **8 and 25** equal-width bins in Army runs.
- **Honest implementation notes** (see `CR_Red_Line_Flow_Explanation.md` , §6.2 below): plot bins use each officer’s **last snapshot** for the binned variable; Cell 11 CIF uses a **simple cumulative proportion** estimator, not full Aalen–Johansen with dynamic risk sets. These are **within-bin empirical summaries**, not Cox-fitted curves.

Layer B — Cell 12: Cox proportional hazards models (inferential)

- **Software:** `scikit-survival` (`sksurv`) — `CoxPHSurvivalAnalysis` , concordance, partial effects.

- **Event framing: Cause-specific Cox** for promotion: officers who attrite are censored at attrition time for the promotion model (and vice versa for attrition models). Cell 12.5 fits **separate** promotion and attrition Cox models and compares coefficients.
- **Competing-risks literature:** Fine–Gray **subdistribution hazard** models are part of the standard toolkit Charles knows; the **current 520 stack implements cause-specific Cox + empirical CIF**, not Fine–Gray regression in code. If manuscript claims Fine–Gray explicitly, confirm implementation first or frame as related estimand family.
- **Typical covariates (example run — CS/CSS, YG 2002–2012):** standardized **own TB ratio** (forward, senior-rater context), standardized **pool minus mean** (forward), **quadratic** terms on both, and **TB × pool interaction**. Documented in [Pertinent_Thoughts.md](#) § Cox Model Results.

Example Cox findings (promotion model, one documented run):

Predictor	Direction / shape	Illustrative magnitude
Own TB ratio (z)	Strong positive on promotion hazard	HR $\approx$ 10.4 per SD
Pool minus mean (z)	Negative on promotion hazard (comparative/competitive)	HR $\approx$ 0.34 per SD
Squared terms (TB, pool)	Significant negative quadratics	Supports curvature / inverted-U shape in hazard scale
TB × pool interaction	Significant	Effect of own performance depends on pool context

All terms significant at conventional levels in that run ($p < 0.001$). Squared terms and interaction justify the **nonlinear** story that the CIF bar panels visualize descriptively.

Conceptual bridge: Cell 11 answers “what does the **binned promotion curve** look like by pool quality?” Cell 12 answers “controlling for own performance and time, what are the **hazard ratios** and **partial effect curves**?” See [CR_AND_HR_FOR_DUMMIES.md](#) for CIF vs hazard vs HR in plain English.

2.4 Key constructs glossary (Army)

Term	Meaning
TB (top block)	Highest OER rating category; often summarized as forward-looking TB ratio or share
Senior rater (SNR)	Senior evaluator whose pool defines peer reference set
Pool minus mean	LOO pool mean TB — peers' average excluding self (Charles's naming; not "minus a fixed cohort mean")
OPM	Own performance measure relative to pool (pipeline has multiple fwd/bwd variants)
Competing risks	Promotion and attrition compete; one precludes the other
CIF	Cumulative incidence of an event by time t
Inverted U	Mobility rises with pool quality, then falls at top pool-quality bins

2.5 Cross-domain replication status (June 2026)

Setting	Agent	Outcome	Pool construct	Inverted-U status
Army officers	CODA	CPT→MAJ promotion (vs attrition)	Senior-rater LOO pool TB	Established — CIF bars + Cox quadratics
College basketball	SCOUT	NBA draft (ever-draft)	LOO teammate pool quality (<code>poolq_loo</code>)	Replicated qualitative shape
Academic tenure	PEER	Asst → Assoc tenure	LOO dept pub intensity / prestige	Preliminary (stage 9, dirty-OK)

Charles's dissertation bar requires **all three settings** in one manuscript (§10). The parallel is **strategic** (nested talent pools, congestion at the top), not a claim that promotion hazards equal draft odds or tenure probabilities.

Primary cross-agent docs: `Publication_Plan.md` §0, `Coda_Summary_For_Scout_and_Vector_Post_Replication.md`, `5-Manuscript/advancement_under_constrained_distinction_dakota_feedback_v03.rtf`.

2.6 Advisor-facing extensions (April 2026)

- **Own-TB stratified CR reruns** (`cr_tb_stratify.py`): same pool-binned inverted-U plots, but restricted to tertiles of **own** TB (low/med/high performers). Tests whether the pool-quality pattern differs by own rating level. Default **off**; see `CR_TB_STRATIFY_Advisor_Three_Panels.md`.
- **525 / UIC work (planned)**: Link pools to divisions, brigades, battalions — which units and senior raters consistently run high-TB pools. Not April 2026 focus; see `525_plans.md`.

2.7 Where to read more (research, not just status)

Document	Content
<code>talent/documents/Publication_Plan.md</code>	Research summary, inverted-U, replication status, mechanism/venues roadmap
<code>talent/documents/CR_AND_HR_FOR_DUMMIES.md</code>	CIF vs Cox vs HR; quadratics; competing risks worked examples
<code>talent/documents/520_PIPELINE_COX_OVERVIEW.md</code>	Full 520 pipeline map (Cells 1–12)
<code>talent/documents/Pertinent_Thoughts.md</code>	Cox result prose, senior-rater pool algorithm notes
<code>talent/documents/advisor_brief_twofold_status.md</code>	Advisor-facing status (525 vs publication)
<code>talent/documents/Coda_Vector_Brief_Army_Evidence_For_Brian_Memo.md</code>	Army bullets for VECTOR / committee
<code>.specstory/history/</code>	Portable chat logs if prose detail is missing from markdown

3. Current status — Army / talent (high level)

Established / working

- `520_pipeline_cox_working.ipynb` : End-to-end pipeline from filtered snapshots through Cox-ready intervals, Cell 10.5 z-score/quadratic/interaction, Cell 11 KM + **competing risks** + **CIF bar panels**, Cell 12 Cox models.
- Inverted-U style figures**: Pool-quality binned CR + final CIF bars (e.g. Q1–Q8) are the advisor-facing signature plot type.
- Run profiles / overrides**: `pipeline_config.py` + overrides such as `pipeline_config_19_1.py` , `pipeline_config_div_name.py` via `pip_config_file` / `expand_run_profile()` .
- Documentation cluster** under `talent/documents/` (see §7).

Recently completed (April 2026 thread — CODA + Charles)

- AWS export overlay**: Files from `talent/Army_AWS_download/TALENT_NET_export_20260421-0802/` copied into `talent/talent_pipeline/`; TB-stratify pieces re-merged afterward.
- Pre-overlay backup**: Full `talent/` tree copied to `obsolete_files/talent_pre_bar_stratify/talent/` .
- Ground-truth policy (Charles decision)**: Everything **outside** `Army_AWS_download/` is authoritative for local work; AWS folder is **comparison snapshot only** until Charles uploads to live Army environment.
- Advisor add-on — own-TB stratified CR reruns**: `cr_tb_stratify.py` + `CR_TB_STRATIFY_CONFIG` + Cell 11 hook. Re-runs `competing_risks` (+ CIF bars when enabled) for low/med/high **own TB** strata (last snapshot per officer). Default `enabled: False` .
- `stratum_method` : `quantile` (equal-N tertiles) vs `equal_width` (equal range on TB/z scale) — implemented in `cr_tb_stratify.py` ; config key in `pipeline_config.py` .

6. **Plot labeling / filenames for TB-stratify runs:** `cr_tb_stratify_title_suffix` on titles/metadata; filename tokens `_tbq_` / `_tbew_` + stratum to avoid overwrites.
7. **Interpretation doc:** `CR_Red_Line_Flow_Explanation.md` — plain English for Cell 11 CR curve construction (last row per officer for plot bins; cumulative not instantaneous).
8. **Pool-size concern documented:** `Pertinent-Thoughts.md` § Senior Rater Pools — `pool_size_snr_*` can exceed 100 under current `(snapshot_date, snr_col)` grouping; likely **definition vs intuition**, not necessarily bad data.

Not done / open

- **Upload to Army AWS live** of the 4-file change set (Charles action).
- **Enable and validate TB stratify on AWS** (`CR_TB_STRATIFY_CONFIG["enabled"] = True` , run Cell 11, check “TB-stratified CR add-on: X created, Y skipped”).
- **Senior rater pool algorithm audit** — Charles believes >100 pool sizes may be impossible for “true” boards; code groups differently (see §6).
- **525 / UIC consistency work** (`525_plans.md`) — still planned, not CODA's April focus.
- **Formal master plan** — explicitly deferred until Charles assigns Master Planner.

4. AWS upload checklist (Charles → live Army coding environment)

Compared byte-for-byte to `TALENT_NET_export_20260421-0802` , these **live** files differ and should be uploaded:

File	Role
<code>520_pipeline_cox_working.ipynb</code>	Cell 11 + TB-stratify hook
<code>pipeline_config.py</code>	<code>CR_TB_STRATIFY_CONFIG</code> (+ rest of config)
<code>cox_plot_helpers.py</code>	TB-stratify title suffix in <code>format_plot_title</code> / config text
<code>cr_tb_stratify.py</code>	New module (not in export folder)

Unchanged vs export (example Charles asked about): `cox_plt_chnge.py` — identical; no upload needed for that file.

After upload: Note in lab notebook: aligned to export **20260421-0802** + TB-stratify add-on + `stratum_method` .

5. Workflow constraints (critical for Master Planner)

Army AWS environment

- Charles often **transcribes by hand** into a locked-down JupyterLab on Army AWS (no casual copy/paste).
- Local Cursor repo is the **authoring** environment; AWS is **deployment**.
- SpecStory history (`.specstory/history/`) is intentionally tracked for **portable agent conversations** — Charles does **not** want SpecStory history ignored in git.

Git / branches (Charles learning git — April 2026)

- Local `main` and `origin/main` had **diverged** around Rivanna workspace + SpecStory `.gitignore` commits vs later talent commits.
- Charles prefers **feature branches** for focused work; avoid mixing Rivanna workspace tweaks with science PRs when possible.
- CODA does **not** own repo-wide git policy — flag for Master Planner if cross-agent commit hygiene matters.

Config reload pattern

- Cell 0 / `reload_pipeline_config()` — changes to `pipeline_config*.py` require reload or kernel restart before Cells 10+.

6. Technical deep dives the Master Planner should know

6.1 Own-TB stratification (advisor add-on)

Intent: Same pool binning and timeline as main CR plots, but **restrict officers** to tertiles of **own** TB (default `z_tb_ratio_fwd_snr` on last interval per `pid_pde`). Produces up to 3× parallel CR + CIF bar outputs per `competing_risks` plot spec.

Docs: `talent/documents/CR_TB_STRATIFY_Advisor_Three_Panels.md`

Enable:

```
CR_TB_STRATIFY_CONFIG["enabled"] = True
# optional: CR_TB_STRATIFY_CONFIG["stratum_method"] = "equal_width"
```

6.2 Cell 11 CR curves — estimand honesty

Key implementation facts (see `CR_Red_Line_Flow_Explanation.md`): - `prepare_plot_data` collapses to `groupby('pid_pde').last()` before binning the plotted variable → **final snapshot** OPM/TB for `plot_group` assignment. - `calculate_cif` uses a **simple cumulative proportion** estimator, **not** full Aalen–Johansen with dynamic risk sets. - **Own-TB strata** (add-on) filter **which officers** enter each rerun; **within-panel pool bins** still come from the plot spec's `variable` / `n_bins`.

Master Planner should ensure manuscript language matches **implemented** estimands, not idealized “snapshot 11 landmark” language unless the data pipeline is changed to that.

6.3 Senior rater pool size > 100

Charles concern: Pool sizes over 100 seem impossible for real senior rater boards.

CODA code reality (`add_cum_oer_metrics_mod_working.py`): - Pools grouped by `[snapshot_date_col, snr_col]` on the long snapshot frame. - `pool_size_*` with `exclude_self=True` reflects effective denominator for pool mean. - Large sizes can occur if `snr_col` is broad (org/UIC/key), many officers share snapshot date, or many rows survive joins — **not** necessarily “one OER board headcount.”

Documented in: `talent/documents/Pertinent-Thoughts.md` (§ Senior Rater Pools, code-linked notes).

Recommended audit when Charles returns: Tabulate top `pool_size_snr_fwd`; for each group report `snr_col`, date, `nunique(pid_pde)`, `yg`, `div_name`; confirm `CELL6_COLUMN_MAPPING` for `snr_col`.

Cross-domain note for Master Planner: SCOUT and PEER have their own pool definitions (`poolq_loo` , dept LOO prestige). Harmonize **language** (“reference peer set at time t”) not necessarily **identical code**.

6.4 Cell 10 warning about missing model TV columns

Message like: `Model-only time-varying missing from snapshot data: ['z_tb_ratio_fwd_snr', ...]`

Usually expected: z-scores, squares, interactions are created in **Cell 10.5**, not on raw snapshots in Cell 10. Run 10.5 before Cell 12 model fitting.

7. CODA document inventory (updated April 2026)

Document	Purpose	Updated this thread?
<code>talent/documents/Pertinent_Thoughts.md</code>	Dissertation ideas + Senior Rater Pools section	Yes
<code>talent/documents/CR_TB_STRATIFY_Advisor_Three_Panels.md</code>	TB-stratify feature guide	Yes (this pass)
<code>talent/documents/CR_Red_Line_Flow_Explanation.md</code>	Cell 11 CR / red-line flow	Yes (created earlier)
<code>talent/documents/README_Talent_Layout_Symlinks_And_AWS_Export.md</code>	Paths, AWS vs ground truth	Yes (this pass)
<code>talent/documents/Coda_Summary_For_Scout_and_Vector_Post_Replication.md</code>	Cross-agent sync (Army side)	Yes (this pass)
<code>talent/documents/CONVERSATION_HANDOFF.md</code>	Technical handoff for new CODA threads	Yes (April section)
<code>talent/documents/advisor_brief_twofold_status.md</code>	Advisor-facing status	Partially current
<code>talent/documents/Agent_Read_First_Coda_Runbook.md</code>	Scout replication runbook (historical)	Reference only
<code>talent/documents/S20_PIPELINE_COX_OVERVIEW.md</code>	Pipeline map	Yes (June 2026) — Cell 11 optional own-TB stratify add-on
<code>talent/documents/Army_to_College_Basketball_Replication_Handoff.md</code>	Long replication design	Historical reference

8. Code map (talent_pipeline essentials)

Path	Role
501_working.ipynb ... 512_oer_int_working.ipynb	Ingest, OER, hierarchies
520_pipeline_cox_working.ipynb	Main Cox + plotting conductor
pipeline_config.py	Base flags, PLOT_CONFIG, CR_TB_STRATIFY_CONFIG, run profile hooks
pipeline_config_19_1.py, pipeline_config_div_name.py, ...	Overrides (from pipeline_config import *)
add_cum_oer_metrics_mod_working.py	Fwd/bwd TB, pool means/sizes/ranks/z
join_oer_to_snapshots_working.py	OER ↔ snapshot join
cox_plot_helpers.py	Plot filenames, CIF bars, titles
cr_tb_stratify.py	TB-stratified CR add-on
py_503_hierarchies.py	UIC / hierarchy helpers

AWS snapshot (not ground truth): talent/Army_AWS_download/TALENT_NET_export_20260421-0802/

9. Sister agents — what CODA knows (read their reports, don’t merge code blindly)

Agent	Report / gameplan (examples)	CODA relationship
SCOUT	sports/documents/SPORTS_DATA_GAMEPLAN.md, Pertinent_Thoughts_Scout.md	Replication setting; inverted-U in draft vs poolq_loo; generative congestion score work
PEER	tenure/documents/TENURE_DATA_GAMEPLAN.md, 5-Manuscript/PEER_Status_Update_for_VECTOR_2026-06-03.md	Setting 3; stage 9 preliminary inverted-U; Cox wiring in progress; not using 520
VECTOR	1-Various_PDE_and_Chat_stuff/5-Manuscript/*.md	Manuscript integration; use Army brief + Scout/PEER status docs

Expected parallel reports for Master Planner: Charles asked each agent for `{AGENT}_report_to_master_planner.md` in `./3-Master_Plan/`. This file is CODA's contribution.

10. Charles answers & open items (Master Planner feed-in)

Recorded 2026-06-08. CODA domain is on **pause** for several items until Charles finalizes the Army/talent lane; re-ask before locking master-plan details.

Answered (use as constraints)

#	Topic	Charles's answer
1	Master Planner agent	Probably brand-new Cursor agent
2	Manuscript structure	One manuscript (all settings)
3	Defense / dissertation bar	All three settings required (not Army-only minimum)
4	Master plan deliverable format	Charles will instruct Master Planner directly
5	Wait for all four agent reports?	Charles will instruct Master Planner directly
11	VECTOR canonical outline	<code>Vector_to_Scout_Tier1_Modeling_Direction.md</code> is not necessarily THE canonical file — wait for VECTOR's report
12	Refresh <code>520_PIPELINES_COX_OVERVIEW.md</code>	Yes — TB-stratify documented as optional Cell 11 add-on (done June 2026)

Deferred — ask again when Charles finalizes CODA domain

- **6** Pool harmonization (language vs code-aligned LOO)
- **7** Pool-size >100 audit vs manuscript disclaimer
- **8** Army AWS upload cadence / TB-stratify upload status
- **9** TB-stratify default on vs off for routine runs
- **10** Priority: 525/UIC vs pool audit

Still open (not answered above)

- **SpecStory + git** policy confirmation
- **Rivanna / workspace** files on science commits vs separate
- **Hard defense date** (if Master Planner needs a timeline anchor)

11. Suggested first tasks for Master Planner (after assignment only)

*Charles asked agents not to master-plan yet; this is **optional seeding** for the assigned planner.*

1. Inventory all four `{AGENT}_report_to_master_planner.md` files + VECTOR manuscript outlines.
 2. Build a **single timeline** (data ready → figure ready → manuscript section ready) per setting.
 3. Resolve **pool definition harmonization** across Army / SCOUT / PEER (shared glossary).
 4. Align **VECTOR** section outline with actual figure availability (Army strongest, tenure preliminary, basketball replicated).
 5. Git hygiene: branch strategy, what belongs on `main`, SpecStory policy.
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12. Agent identity note

This report was authored by **CODA** in the Cursor session covering Army talent work with Charles (TB stratify, AWS overlay, CR interpretation, pool-size concerns, Pertinent Thoughts updates). If Charles intended a different agent name for this thread, relabel this file accordingly.

End CODA handoff. Do not begin master planning until Charles assigns the Master Planner agent.